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Hiring Team  
Vaccines Biostatistical Sciences  
Sanofi, Marcy-l'Étoile, France

**Subject: Application for Scientific Data Analyst - Vaccines Biostatistical Sciences**

Dear Hiring Team,

I am applying for the Scientific Data Analyst position within Sanofi Vaccines Biostatistical Sciences. I hold a PhD in experimental physics and have several years of experience developing Python/C++ workflows for statistical modelling, data manipulation, validation, inference, and monitoring on large, noisy datasets. Combining modern open-source programming, AI/ML, and rigorous evidence generation in clinical development strongly matches the work I want to pursue in industry.

My research has repeatedly required end-to-end analytical workflows where the system state was only indirectly observable. I implemented probabilistic and maximum-likelihood models under background, finite resolution, selection effects, and dataset mismatch. I also built CatBoost pipelines, contributed covariance-aware validation and pseudoexperiments, and developed monitoring that converted discrepancies into calibration or software corrective actions.

I am accustomed to Linux-based shared codebases using Python, C++, Git, modular programming, collaborative review, reproducible configurations, and automation. I have produced statistical summaries, diagnostic figures, documented methods, and peer-reviewed results for international collaborations. This provides a strong foundation for reusable analytical tools and close work with biostatisticians and other scientists.

I want to be transparent about the transition. I have not yet worked professionally with clinical-trial data, SDTM/ADaM, Statistical Analysis Plans, submission outputs, SAS, or Pharmaverse. I would address these gaps immediately through a public CDISC pilot dataset, analysis outputs in R and Python, traceability and automated QC checks, and study of the clinical-analysis lifecycle. A short project cannot replace regulated-industry experience, but my statistical and software foundation would let me learn the technical concepts quickly while adopting Sanofi's standards with the team.

My move from particle physics toward industry is deliberate: I am most motivated by solving difficult real-world analytical problems, making models trustworthy, and delivering reusable methods. Applying these strengths to vaccine clinical development would combine technical depth with direct impact on human health. I would welcome the opportunity to discuss how my statistical analysis, validation, automation, and multidisciplinary collaboration could contribute to Sanofi's statistical programming practice.

Yours sincerely,

Yingrui Hou